

0625 | PAPER 4

TOPIC 05

Nuclear Physics

39 topical answers • 2021–2025

Calculations, final values and examiner-keyword summaries

Answers are independently condensed from the official marking requirements. They are not a reproduced Cambridge mark scheme. For drawings and graph lines, compare the stated method with the original question figure.

Topic contents

Subtopic	Questions
5.1 Atoms, nuclei and isotopes	8
5.2 Alpha, beta and gamma radiation	6
5.3 Half-life	20
5.4 Nuclear fission and fusion	4
5.5 Radioactivity - mixed	1

Core formulas

- mass number = protons + neutrons
- activity falls randomly with time
- after n half-lives: remaining fraction = $(\frac{1}{2})^n$

5.1 Atoms, nuclei and isotopes

Q288 • May/June 2021 Paper 41 Question 9

5.1 Atoms, nuclei and isotopes

(a) 2

 1H and 1H and in this order

(b)(i) joining together of (small / H) nuclei

to produce a bigger nucleus / He nucleus or with the release of energy

(b)(ii) ($21\text{H} + 31\text{H} \rightarrow 01\text{n}$)

(+) 42 (....)

He or α seen

(c) Any two valid point(s):

- geothermal (energy)
- tidal (energy)
- nuclear (energy)

Q289 • Oct/Nov 2021 Paper 43 Question 11

5.1 Atoms, nuclei and isotopes

(a)(i) background radiation; or any reasonable specific source of background radiation e.g. cosmic rays, the sun, space, building materials

earth, rocks, radon gas, student etc.

(a)(ii) (radioactive decay is a) random (process)

(b)

U: proton no 92 and nucleon number 238

Th: proton number 90 and nucleon number 234

 α : proton number 2 and nucleon number 4

(c) 11

three half lives or evidence of multiplying half-life by 3

Q290 • Feb/March 2022 Paper 42 Question 10

5.1 Atoms, nuclei and isotopes

(a) 241

 $\rightarrow 237\ 4$ $95\text{Am}\ 93\text{Np} + 2\ \alpha$

Np nucleon number correct for Np

 93Np proton number correct for Np $+ 42\ \alpha$ alpha notation correct

(b)(i) alpha (particles emitted from americium)

move close to / hit molecules in the air (between the metal plates)

removing electrons (out of the molecules)

(b)(ii) Any two valid point(s):

- alpha not penetrating / short range and alpha (particles) stopped by smoke particles
- alpha (particles) more highly ionising (than gamma) and ionise air more easily
- range of alpha particles is short / alpha is not penetrating and alpha less harmful (to humans)

Q291 • May/June 2022 Paper 41 Question 10

5.1 Atoms, nuclei and isotopes

(a)(i) same number of protons / both have one proton

(a)(ii) it / hydrogen-3 / (13H) has one more neutron

different number of neutrons / nucleons

(b)(i) (high temperature produces) high (kinetic) energy / momentum / speed / ability to do large quantity of work

they repel each other

are positively charged / have like charges or need to come close together

(b)(ii) 4

or 42 He or 42α
 $2X$
 $0n$ and no other particle

Q292 • May/June 2022 Paper 42 Question 10

5.1 Atoms, nuclei and isotopes

(a)

output of X output of Y

1

0

all column X correct 1, 1, 0, 0

first 2 rows of column Y correct 0, 0

last 2 rows of column Y correct 1, 0

(b) high humidity and dark(ness)**(c) relay**

low voltage output (of NOR gate/gate Y)

or small current (in relay coil)

large(r) current provided (by relay)

or large(r) voltage provided (by relay)

Q293 • Feb/March 2023 Paper 42 Question 9

5.1 Atoms, nuclei and isotopes

(a)(i) large unstable nucleus or neutrons hit nucleus or neutrons are released (from nucleus)

(large) nucleus splits (into smaller nuclei)

(large) release of energy

(a)(ii) advantage - one from:

- Continuous supply of energy
- not affected by the weather or not affected by wind strength
- produces large amounts of energy

disadvantage - one from:

- resources finite / not renewable
- cost / difficulty of building / cost / difficulty of decommissioning
- danger if any leak of radiation
- produces hazardous / dangerous waste; or difficulty of storage of used radioactive material stored for a long time

(b) 2 $+ 2^1_1\text{H} \rightarrow 3^2_2\text{He} + 0^1_0\text{n}$

1H

LHS correct

2 He on RHS

0 n on RHS

Q294 • May/June 2023 Paper 43 Question 8

5.1 Atoms, nuclei and isotopes

(a)(i) (neutron) becomes proton and electron or (neutron) becomes proton and β particle**(a)(ii) charge on neutron = 0 or total charge on products = 0**

charge on proton = +1 and charge on electron = -1

(b) **$6\text{ Rn} \rightarrow 84\text{ Po} + 2\alpha$**

(212 208

Any two valid point(s):

- proton numbers balance
- nucleon numbers balance

4 4

- 2 α or 2He seen

Q295 • Oct/Nov 2023 Paper 42 Question 8

5.1 Atoms, nuclei and isotopes

(a) (92 is) the proton number / number of protons (in the nucleus) / atomic number

(235 is) the nucleon number / number of nucleons (in the nucleus) / mass number

(b)(i) (nuclear) fission**(b)(ii) nucleus converted to (more stable) nuclei with smaller total mass**

mass (difference) is released / converted as (kinetic) energy (of products) / thermal energy

(c)(i) Any three valid point(s):

- (thermal energy) used to heat / boil (cold) water or make steam
- steam is at high pressure
- steam drives a turbine
- turbine (connected to and) drives a generator
- turbine moves a coil in a magnetic field

(c)(ii) advantage - Any one valid point(s):

- (much) small(er) amount of fuel needed (to produce same amount of energy)
- no greenhouse gases produced or low carbon dioxide emissions
- no air pollution (when operating normally)

disadvantage - any one from

- danger if any leak of radiation
- produces hazardous / dangerous / toxic waste; or difficulty of storage of used radioactive material must be stored for a long time
- expensive to build or decommission nuclear power plant or store nuclear waste

5.2 Alpha, beta and gamma radiation**Q296 • Oct/Nov 2021 Paper 41 Question 9**

5.2 Alpha, beta and gamma radiation

(a) (very small) nucleus and surrounded by electrons (in orbit / shells)**2 protons or 92 electrons or number of protons = number of electrons**

protons and neutrons in nucleus

143 neutrons

(b) (uranium-238 has) three more neutrons (in nucleus)**(c) 94**

(38) (E)

38 (E)

(d)(i) 55**(d)(ii) 140****Q297 • Oct/Nov 2021 Paper 42 Question 11**

5.2 Alpha, beta and gamma radiation

(a) (very small) nucleus and (surrounded by) electrons (in orbit / shells)

neutrons and protons in nucleus

4 electrons (in atom) or number of electrons = number of protons

4 neutrons (in nucleus)

(b) 135

on left

Cs on left

Ba on right

+ β on right or - β on left**Q298 • May/June 2024 Paper 41 Question 8**

5.2 Alpha, beta and gamma radiation

(a)(i) 0

- 1

β

.....Pb

.....

2Pb**(a)(ii) γ -emission / it consists of waves / rays or γ -emission has no mass / charge****(a)(iii) (it contains) too many / excess of neutrons or (nucleus is) too heavy****(b) smooth curve (through magnetic field) and labelled β** path towards bottom of page and no upward component and labelled β (continuation of beam along) horizontal line through magnetic field and labelled γ **Q299 • May/June 2024 Paper 42 Question 9**

5.2 Alpha, beta and gamma radiation

(a)(i) Any two valid point(s):

- reduce exposure time and low(er) amount of radiation absorbed
- increase distance between source and hospital staff and lower amount of radiation reaches staff
- use of shielding (e.g. walls, lead etc.) and radiation absorbed by shielding / cannot penetrate through shielding
- use isotopes with short half-life and lower amount of radiation emitted from patient / radiation (above background) emitted for a shorter period of time
- use of film badge / dosimeter and manage individuals exposure
- restrict pregnant staff / patient in hospital and radiation may harm foetus

(a)(ii) high ionisation (within body)

radiation would not reach detector (outside body)

(b) 24 24 0 $^{11}\text{Na} \rightarrow ^{12}\text{Mg} + ^{-1}\beta$ ^{11}Na on LHS

• 1

 β on RHS

Mg on RHS

Q300 • Oct/Nov 2024 Paper 43 Question 8

5.2 Alpha, beta and gamma radiation

(a)(i) fission is splitting of nuclei or fusion is the joining of nuclei**(a)(ii) 10 1 7 4** $5\text{B} + 0\text{n} \rightarrow 3\text{Li} + 2\alpha$

B: proton number 5 and nucleon number 10

Li: proton number 3 and nucleon number 7

 α : proton number 2 and nucleon number 4**(b) alpha is less penetrating / has shorter range (than gamma)**alpha more easily absorbed / stopped by cancer / tumour cells; or won't travel beyond cancer cells
other / healthy cells

alpha highly ionising (than gamma)

will destroy / damage cancer cells more easily

(c) Any one valid point(s):

- reduce exposure time
- increase distance between source and living tissue; or stand in another room whilst radiation is emitted
- use shielding; or wear a lead apron

Q301 • Feb/March 2025 Paper 42 Question 9

5.2 Alpha, beta and gamma radiation

(a)(i) Key physics: isotopes forms element same number protons but different neutrons nucleus

(isotopes are) the same element with a different number of neutrons

(a)(ii)

particle number in each location

atom of strontium

electron 38 outside nucleus

neutron
 proton 38 Inside nucleus
 particle number in each location
 atom of strontium
 38 outside nucleus
 neutron 52 Inside nucleus
 38 Inside nucleus

(b)(i) (there is a) different count rate with different thicknesses of metal; or number of β -particles detected varies with thickness

(beta (β) particles) can penetrate thin / 0.75mm metal or (beta (β) particles) are stopped by thick metal

(b)(ii) (approximate) percentage of source remaining (after 15 years) stated and in range 63-70%

(approximate) percentage of sample lost (after 15 years) stated and in range 30-37%

Any one valid point(s):

- count rate / activity (too) low (to detect differences in thickness)
- detector needs high activity (to detect differences in thickness)
- count rate / activity too close to background
- less difference in activity for different thicknesses

5.3 Half-life

Q302 • Feb/March 2021 Paper 42 Question 10

5.3 Half-life

(a) 2

+2

(b) 90 90 0

$\text{Sr} \rightarrow \text{Y} + \beta$

38 39 -1

nucleon numbers 90 on both sides of equation

Sr and proton number 38 on left and Y and proton number 39 on right

β (to right of arrow)

- 1

(c) (original mass = 4 / 9.2 =) 37 mg

2 half-lives

Q303 • May/June 2021 Paper 42 Question 10

5.3 Half-life

(a) $V = IR$ in any form or $(R =) V / I$

$(R = 9.2 / 0.004 =) 2300 \Omega$

(b) (much) greater current in lamp or lamp activated / lights / glows / gets brighter

resistance of thermistor / component / K reduced (compared to value at (very) low temperature)

voltage / p.d. of point X / across R increases

(larger) current in lamp

(c) thermistor

Q304 • May/June 2021 Paper 43 Question 10

5.3 Half-life

(a)(i) curve bending downwards while in magnetic field (and labelled α)

(a)(ii) curve bending in opposite direction from α while in magnetic field or

up the page if no curve shown for α in (a)(i) (and labelled β)

greater curvature for β than for α

(a)(iii) line passing straight through magnetic field (and labelled γ)

(b) Any two valid point(s):

- stand behind shielding provided / wall / as far away as possible
- store in lead-lined boxes
- limit exposure time / (monitoring exposure) with film badge

- do not pregnant staff to work

(c)(i) 131

53 l

(c)(ii) Any two valid point(s):

- γ can be detected outside body
- needs long enough half-life to be detected / reach part of the body required
- needs short enough half-life to soon have very little activity
- gamma weakly ionising or pass out of body without harm

Q305 • May/June 2022 Paper 43 Question 10

5.3 Half-life

(a) background radiation (present in values in Table 10.1)

(background radiation) is removed (before plotting) or (background radiation) not present in the graph values

(b) $70 \leq \text{half-life} \leq 76$ (s)

evidence of at least one pair of values for count rate halving taken from graph

(c) 234 234 0

Pa $\rightarrow$ U + β

91 92 -1

92U on RHS

234U

+ β on RHS

• 1

Q306 • Oct/Nov 2022 Paper 41 Question 9

5.3 Half-life

(a) they all have the same number of neutrons / nucleons or they are all identical

(b)(i)

(number of protons =) 80

(number of neutrons =) 118

(b)(ii) 19 counts / minute $\leq$ count rate $\leq$ 21 counts / minute

(b)(iii) 2.4 days $\leq t \leq$ 2.9 days

count rate from line - background count e.g. 390 - 20

answer from first C1 mark divided by 2 e.g. 370 / 2 or 185

background count + answer from second C1 mark e.g. 20 + 370 / 2 or 20 + 185 or 205

Q307 • Oct/Nov 2022 Paper 42 Question 10

5.3 Half-life

(a)(i) (proton number) 2

(nucleon number) 4

(a)(ii) 3.2×10^{-19} (C)

230 230 0

$88\text{Ra} \rightarrow 89\text{Ac} + -1\beta$

(b)

Any two valid point(s):

- nucleon numbers 230 on left and 230 on right
- Ra and proton number 88 on left and Ac and proton number 89 on right
- 1

β .

(c) (mass =) 1.2×10^{-12} g

3 half-lives or $9.6 \times 10^{-12} / 8$ or $9.6 \times 10^{-12} / 23$

Q308 • Oct/Nov 2022 Paper 43 Question 10

5.3 Half-life

(a) α -particles are short and thick / β -particles are long and thin

Any two valid point(s):

- α -particles are more ionising / β -particles are less ionising

- α -particles are less penetrating; or have shorter range / β -particles are more penetrating
- α -particles have more energy / β -particles have less energy

(b)(i) (element with) same number of protons

(element with) different number of neutrons

(b)(ii) 24 24 0

11 12 -1

Na on left with correct proton and nucleon number

 β on right with correct proton and nucleon number

Mg on right with correct proton and nucleon number

(b)(iii) half life (of Na 24) long enough (to detection of leaks)

negligible amount (of Na 24) remains in liquid after a few days

(so) less hazardous (to human health)

decays to something stable/magnesium (is stable)

and (so) less hazardous (to human health)

Q309 • May/June 2023 Paper 41 Question 9

5.3 Half-life

(a) (number of neutrons =) 7

Any one valid point(s):

- number of electrons = number of protons
- white dots are protons / there are 5 protons
- grey dots are neutrons
- (number of neutrons) = 12 - 5

(b)(i) (X2 has) one more proton more and one fewer neutron (than X1) or (X2 has) 6 protons and 6 neutrons

(X2 has) one neutron fewer / one more proton (than X1)

or (X2 has) 6 protons / 6 neutrons

(b)(ii) (X2) has fewer (excess) neutrons (in its nucleus)**(c)(i) time (taken)**

for number of (radioactive) nuclei / atoms (in a sample of X1) to halve or for rate of decay to halve

(c)(ii) large number of particles produced in short time; or high / large decay rate**Q310 • May/June 2023 Paper 42 Question 9**

5.3 Half-life

(a) α - no. of neutrons 2 β - no. of protons 0 and charge -1.6×10^{-19} γ - no. of neutrons 0 and charge 0 and (very) thick concrete / thick lead**(b) (the nucleus has) one less neutron and one more proton****(c) 95 (counts / min)**

initial count rate due to source = 550 - 30 (counts / min) or 520 seen

(75 min $\Rightarrow$) 3 half-lives or (count rate $\Rightarrow$) 1 / 8 (of initial count rate)final count rate due to source = (520 / 8 $\Rightarrow$) 65**(d) Any two valid point(s):**

- limit time of exposure
- store sources in lead boxes
- keep distance from sources
- avoid contact or use tongs or wear gloves

Q311 • Oct/Nov 2023 Paper 41 Question 9

5.3 Half-life

(a)(i) (americium-241 has) one neutron fewer (in the nucleus)**(a)(ii) (different) number of protons (in nucleus)**

(different) number of neutrons (in nucleus)

(b)(i)

5 Am $\rightarrow$ 237

3 Np + 2 a

Any two valid point(s):

5 Am

Np

2a

(b)(ii) (α -particles have) more kinetic energy (than β -particles)

(α -particles have) more charge (than β -particles)

(b)(iii) Low(er) (initial) activity or Few emissions per unit time

so smoke detectors are not hazardous to humans; or so disposal of old detectors is cheap / easy

Q312 • Oct/Nov 2023 Paper 43 Question 9

5.3 Half-life

(a)(i) alpha

highly ionising

or not (very) penetrating

any value between 10-500 years

(a)(ii) beta

absorption depends on thickness (of aluminium)

any number of years

(b) too many neutrons

decay reduces number of neutrons

(c) Any two valid point(s):

- radon gas (in the air)
- rocks or buildings
- food or drink
- cosmic rays

Q313 • Feb/March 2024 Paper 42 Question 9

5.3 Half-life

(a)(i) most of the atom is empty space

(a)(ii) Any two valid point(s):

1 the nucleus is very small

2 mass of gold nucleus is much greater than mass of alpha particle

3 the nucleus is positively charged

corresponding explanation to conclusion:

1 not many alpha particles pass close to the nucleus /

2 large force between alpha and nucleus (has bigger effect on small mass of alpha)

3 alpha particles are positively charged, and force is repulsive

(b)(i) (in the nucleus a) neutron is changed into a proton (and an electron which is the emitted β -particle)

(b)(ii) 22 μ g

(87 years is) three half -lives or $25 / 8$ or $87 / 29 = 3$ (half lives)

1 / 8th (of the strontium remains) or $25 / 8$ (decays) or 3.125 seen

Q314 • May/June 2024 Paper 43 Question 10

5.3 Half-life

(a)(i) radiation (always) present in the environment or radiation from natural sources

(a)(ii) radon gas or rocks / buildings or cosmic rays or food / drink

(a)(iii) 356 counts / min

(corrected count rate =) count rate - background count or 382-26

(b)

beta or gamma

alpha would be stopped by soil and not reach detector alpha and beta stopped by soil before reaching detector

or beta has sufficient range to be detected above or

ground

gamma would pass through intact pipe as well cracked gamma is only radiation that will reach detector through pipe so no difference detected; or beta stopped by soil
intact pipe so variation between piped water and pipe
escaped water can be detected

(c)(i) (very short half-life) doesn't time for detection before activity has dropped to too low a level; or doesn't for

sufficient build up at the leak to detect difference in rate

(c)(ii) (very long half-life) contaminates water supply

Q315 • Oct/Nov 2024 Paper 41 Question 8

5.3 Half-life

**(a) 6 electrons and 6 protons (i.e. 6 × and 6)
neutrons (i.e. 8)**

protons and neutrons in nucleus and electrons orbiting nucleus

(b) (carbon) has one more neutron or nitrogen has one fewer neutron

(carbon) has one fewer proton / electron or nitrogen has one more proton / electron

(c)(i) electron

(c)(ii) a neutron changes into a proton (and electron)

(d) 17 000 years

$1.2 \times 10^{11} / 9.6 \times 10^{11}$ or $1 / 8$ or one halving seen e.g. $9.6 \times 10^{11} \div 2$

3 (half-lives) or $1 / 8 \times 9.6 \times 10^{11} = 1.2 \times 10^{11}$

Q316 • Oct/Nov 2024 Paper 42 Question 10

5.3 Half-life

(a) (nucleus of carbon-14 contains) more neutrons or (nucleus of) carbon-12 has fewer neutrons

Any one valid point(s):

- (atom / nucleus of carbon-14) is heavier
- (atom / nucleus of carbon-14 is) not stable
- (nucleus of carbon-14 contains) two more neutrons

(b)(i) Every 5700 years the (remaining) carbon-14 decreases by half or amount of C-14 halved every half life

amount of carbon-14 halves in 5700 years or amount of carbon-14 decreases at a decreasing rate (with time)

(b)(ii) 11 000 (years ago)

$2 \times$ half-life have elapsed

25% of C-14 at time of death is still present in tree or 75% of C-14 has decayed.

(c) medical tracers or medical imaging or medical diagnosis

Any one valid point(s):

- keep dose low
- doesn't stay in body too long
- less damage (to body) or less harmful (to humans)

Q317 • May/June 2025 Paper 41 Question 9

5.3 Half-life

(a)(i) (stable isotope) has fewer neutrons and radioactive isotopes (usually) have an excess of neutrons

(stable isotope) has fewer neutrons and radioactive isotopes are too heavy

Any one valid point(s):

- (stable isotope) has fewer neutrons
- radioactive isotopes (usually) have more neutrons
- radioactive isotopes are too heavy

(a)(ii) 90

39Y

β

- 1

(a)(iii) ionising radiation is harmful (to humans) or

beta particles are ionising and harmful (to humans)

Any one valid point(s):

- radiation is / beta particles are harmful
- beta particles ionise
- it ionises and is harmful

(b)(i) $72 \leq \text{half-life} \leq 76$ (h)

braille: 80 h (mark based on candidate choice of halving)

evidence of count rate halved e.g. $48 \div 2 = 24$

evidence on graph or in working that Fig. 9.1 is used to find time for count rate to halve

(b)(ii) Any one valid point(s):

- table includes background radiation
- graph does not have background count rate
- graph has corrected count rate

Q318 • May/June 2025 Paper 42 Question 9

5.3 Half-life

(a)(i) radiation (always) present in the environment or radiation from natural sources

(a)(ii) Any one valid point(s):

- radon gas
- rocks or buildings
- food or drink
- cosmic rays

(b)(i) 219

^{86}Rn

4α

(b)(ii) alpha (particles) would be absorbed / stopped by the skin

(c)(i) gamma / radiation needs to pass out of the body (to detector)

(c)(ii) after a few days / some time, little radiation is emitted

Q319 • May/June 2025 Paper 43 Question 11

5.3 Half-life

(a) time taken for half the nuclei (in any sample) to decay

(b)(i) 234(U)

^{92}U

$+ -1 \beta$

(b)(ii) neutron changes to a proton (plus an electron)

(c) (α particles have) greater kinetic energy (than β particles)

(α particles have) greater charge (than β particles)

Q320 • Oct/Nov 2025 Paper 42 Question 9

5.3 Half-life

(a)(i) curve bending upwards while in magnetic field and labelled α

(a)(ii) curve bending downwards while in magnetic field and labelled β

smaller radius of curvature for β than for α

(a)(iii) line passing straight through the magnetic field and labelled γ

(b)(i) time taken for half the nuclei of a sample to decay

(b)(ii) R

half-life not too short; or half-life is suitable so that source does not need to be replaced often

so count rate changes with thickness

γ -rays would not be stopped at all and α -particles would be absorbed by any thickness of aluminium

Q321 • Oct/Nov 2025 Paper 43 Question 8

5.3 Half-life

(a)(i) most of the atom is empty space

nucleus is (very) much smaller than the atom

(a)(ii) nucleus must have a positive charge

(the positive) charge is concentrated at the centre of the atom

most of the mass (of the atom) is concentrated in the nucleus

(b)(i) P

alpha particles have greater mass so are deflected less (than the beta particles)

(b)(ii) upwards

alpha particles are positively charged so attracted towards negatively charged plate; or direction of field is direction of force

on positive charge

(c) 1.9×10^{-2} g or 1.9×10^{-5} kg

3 half-lives or $2.4 \times 10^{-3} \times 23$ or $1/8$

5.4 Nuclear fission and fusion

Q322 • May/June 2021 Paper 43 Question 4

5.4 Nuclear fission and fusion

(a)(i) Energy transferred when 1 kg / unit mass of a substance freezes or melts

Energy transferred when a substance freezes/melts/changes state

(a)(ii) cup containing mixture of ice and water

mixture of ice and water will remain at 0°C until all ice is melted (but

temperature of water at 0°C rises) or reverse argument

or energy needed for change of state so temperature doesn't rise until this has taken place

(b)(i) in evaporation more - energetic / faster moving molecules / molecules with high(er) kinetic energy escape (from surface)

low(er) energy / slow molecules remain or so remaining liquid is cooler

thermal energy is taken from person to liquid (so person cools down)

(b)(ii) (great(er) / fast(er) evaporation of sweat as) wind blows fast moving

molecules away or molecules do not re-enter the liquid

Q323 • Oct/Nov 2021 Paper 41 Question 2

5.4 Nuclear fission and fusion

(a) (quantity of thermal) energy or energy (to increase temperature) or energy (transferred by heating)

energy to increase temperature (of an object) per degree Celsius 1°C

(b)(i) (internal energy) depends on kinetic energy (of molecules)

kinetic energy (of molecules) decreases or potential energy (of molecules) decreases

(b)(ii) ($\Delta E =$) $mc\Delta T$ in any form or $0.24 \times 4200 \times 17$

$1.7 \times 10^4 \text{ J}$

(c)(i) k.e. of molecules / (thermal) energy absorbed (from water / surroundings); or energy absorbed from (cooling) water

supplies latent heat or energy used to overcome intermolecular forces / to break bonds

(c)(ii) any determination of mass

determine change in mass (of ice) / increase in mass of water or dry the ice or ensure water is at 0°C / equilibrium is established or insulate the beaker

use ($l =$) E / m in any form

Q324 • May/June 2022 Paper 43 Question 5

5.4 Nuclear fission and fusion

(a) energy required to raise the temperature of 1 kg / 1 g / unit mass of a substance by 1°C / unit temperature

energy required to raise the temperature of a substance by 1°C

(b)(i) 0.50 kg

$\rho = m/V$ in any form

(b)(ii) 190 000 J / $1.9 \times 10^5 \text{ J}$ / 190 kJ

($E =$) $mc\Delta T$ in any form

($E =$) mL in any form

Use of $c = 4200 \text{ (J / kg }^\circ\text{C)}$ and $\Delta T = 5$

Use of $c = 2100$ and $\Delta T = 18$

Q325 • Oct/Nov 2025 Paper 41 Question 9

5.4 Nuclear fission and fusion

(a) Any three valid point(s):

- nucleus containing protons and neutrons
- protons (in nucleus)
- 2 neutrons (in nucleus)
- 2 electrons outside the nucleus
- Protons are positive (charges) or electrons are negative (charges)

(b)(i) hydrogen

(hydrogen) nuclei join (together)

(b)(ii) Any two valid point(s):

- infrared
- visible
- ultraviolet

(c) • (the star) expands

Any one valid point(s):

- (smaller star) forms a red giant
- (more massive star) forms a red supergiant

5.5 Radioactivity - mixed**Q326 • May/June 2021 Paper 42 Question 8**

5.5 Radioactivity - mixed

(a)(i) α in Box 4 / towards bottom of page γ in Box 3 / no deflection**(a)(ii) α in Box 1 / into page** γ in Box 3 / no deflection**(b)(i) clockwise rotation arrow on diagram**

force on L wire up / up arrow on L wire labelled force on diagram

force on RH wire down / down arrow on R wire labelled force on diagram

(b)(ii) none / zero (moment)**(b)(iii) current in coil reverses or changes direction**

force(s) (on wires in new positions) still up on L or down on R